Current–Wave–Resolution Cascade (CWRC)

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A Re-expression of CRF Collective Dynamics in Wave Language

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Abstract

The Current–Wave–Resolution Cascade (CWRC) is not a new framework. It is the existing δ-chain expressed in wave language: the base current J is the instability gradient $G = \nabla \Pr(s|C)$; the wave field Φ is $D_{\mathrm{KL}}(C\|P)$ propagating through space; and the pattern modes Ψ_i are the accumulated $\Pr(s)$ distributions of individual Minds identified in CRF Sessions 4–5.

This document maps each CWRC element to its δ-chain origin, identifies what is already derived, what is hypothesis, and what is genuinely new. The one genuinely new contribution is the *Internal Dimensional Volume* (IDV) — the intuition that two nodes of identical external size can contain radically different effective dimensionality. This is formalized here for the first time.

Contents

1 The Mapping: CWRC to δ-Chain

Every element of CWRC has an exact counterpart already derived in CRF. The table below is the canonical translation.

CWRC to CRF Canonical Mapping

CWRC term	CRF canonical	Source
J — base current	$G = \nabla \Pr(s\|C)$, instability gradient	Session 6
Φ — wave field	$D_{\mathrm{KL}}(C(x,t)\|P)$, the gap propagating	EIC Bridge
Ψ_i — pattern mode	P_M = accumulated $\Pr(s)$ of Mind M	CRF Identity
Selective coupling $e^{-\|\Psi_i - \Psi_j\|}$	$\Pr(\text{sync}) \propto e^{-\alpha D_{\mathrm{KL}}}$	CRF Synchronicity
Wave lock / resonance	Inevitability: $D_{\mathrm{KL}} \to 0 \Rightarrow$ events feel inevitable	CRF Synchronicity
$J \to \Phi \to \Psi_i$ cascade	$\delta \to G \to D_{\mathrm{KL}} \to P_M$	Sessions 1–5

Status of CWRC

CWRC is a *re-expression*, not a new derivation. All elements were already derived from δ. Its value is pedagogical: wave language makes the collective dynamics of CRF visually and intuitively accessible. It should appear as a *Supplemental Note* to CRF, not as an independent framework.

2 What CWRC Adds That Is Genuinely New

Two things emerged from the CWRC exploration that are not yet formalized in any CRF document:

2.1 The Phase Field Φ as Spatial Wave

CRF Synchronicity derived collective dynamics for a *network* of Minds. CWRC goes one step further by treating D_{KL} as a *spatial field* that propagates and reflects:

$$\Phi_i(t) = D_{\mathrm{KL}}(C_i \| P_{\text{shared}})$$

The wave reflection term is:

$$\Phi_i^{\text{reflect}}(t) = \sum_{j \in \mathcal{N}(i)} D_{\mathrm{KL}}(P_i \| P_j) \cdot \sin(\Phi_j - \Phi_i)$$

This is the Kuramoto model applied to the CRF instability field — a derivation confirmed in the image included with this document (Kuramoto as special case of CRF Synchronicity Formalism).

Kuramoto Derivation from CRF

The Kuramoto equation for oscillator i:

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i)$$

maps to CRF via:

$$\theta_i \leftrightarrow \Phi_i, \quad \omega_i \leftrightarrow \Omega_i = D_{\mathrm{KL}}(C_i \| P), \quad K \leftrightarrow \mu \text{ (resonance operator, Sessions 7–8)}$$

$$\sin(\theta_j - \theta_i) \text{ emerges from } \Pr(\text{sync}) \propto e^{-D_{\mathrm{KL}}(C_i \| C_j)} \text{ when } P \text{ is periodic.}$$

Kuramoto measures *how* Minds sync. CRF explains *why* they must sync (toward $G = 0$, Axiom 11). Status: **Derived**.

2.2 Pattern Mode Individuality

Ψ_i (pattern mode) $= P_{M_i}$ (accumulated distribution of Mind i). The selective coupling $\exp(-|\Psi_i - \Psi_j|/\sigma)$ is the CRF synchronicity formula $\Pr(\text{sync}) \propto e^{-\alpha D_{\mathrm{KL}}(C_i \| C_j)}$ with σ as a scale parameter.

Nodes with similar Ψ (similar accumulated history) couple strongly. Nodes with different Ψ decouple. This is *speciation* in CRF language:

$$\text{Species} \equiv \text{cluster of nodes with low pairwise } D_{\mathrm{KL}}$$

3 The Genuinely New Result: Internal Dimensional Volume

This is the only contribution in this document that does not exist elsewhere in CRF.

3.1 The Intuition

Two nodes can have identical external coordinates (same position in X-space) but radically different effective dimensionality. This is the "glass marble" insight: two marbles the same size, but one contains $1\,\mathrm{m}^2$ inside and another contains $10^6\,\mathrm{m}^2$ inside.

In CRF: two nodes with the same X-position can have different numbers of accessible states (different effective d_s), determined entirely by their accumulated $\Pr(s)$ history and D_{KL} structure.

3.2 Formal Definition

Internal Dimensional Volume (IDV)

For node i with neighbors $\mathcal{N}(i)$, define:

$$V_i^{\text{int}} = \underbrace{H(P_i)}_{\text{config richness}} \times \underbrace{\left(1 + \mathrm{Var}_{j\in\mathcal{N}(i)}[\cos(\Phi_j - \Phi_i)]\right)}_{\text{phase diversity}} \times \underbrace{\left(1 + |\nabla\Psi|_i\right)}_{\text{speciation gradient}} \times \underbrace{\ln(1 + \text{mass}_i)}_{\text{memory depth}}$$

where $H(P_i) = -\sum_k P_i(k)\ln P_i(k)$ is the entropy of node i's distribution.
Local effective dimension:

$$d_i^{\text{int}} = \ln(1 + V_i^{\text{int}})$$

Global indicator of Internal Dimensional Inflation (IDI):

$$\overline{V^{\text{int}}}(t) = \frac{1}{N}\sum_{i=1}^{N} V_i^{\text{int}}(t)$$

If $\overline{V^{\text{int}}}$ increases while global $d_s \approx 3$, the system is *internally expanding* while externally stable.

3.3 Grounding in the δ-Chain

Each factor in IDV maps to a CRF quantity:

IDV factor	**CRF derivation**
$H(P_i)$	Entropy of C (constraint distribution). From Session 4: $C \equiv \Pr(s\|C)$, entropy $H = -\sum \Pr(s)\ln \Pr(s)$.
Phase diversity	Variance of $\cos(\Phi_j - \Phi_i)$ measures how synchronised the local instability wave is. Low variance = near Axiom 11 ($G \to 0$).
Speciation gradient $\|\nabla\Psi\|_i$	Difference in accumulated P_M between neighbours. High gradient = speciation boundary = region where distinct matter types meet.
$\ln(1 + \text{mass}_i)$	$\Omega_i = \text{mass}_i - 1 = D_{\text{KL}}(C_i\|\|P_{\text{bg}})$, the local CRF field (EIC Bridge). Log ensures IDV grows sublinearly with age.

Open: Formal Proof of IDV

The IDV formula is constructed to be consistent with the δ-chain but has not been derived from axioms alone. Specifically: (1) the product form is an ansatz, not forced; (2) the relationship between d_i^{int} and the global d_s estimator (random-walk return) has not been proved. Status: **Hypothesis** pending formal derivation.

4 Connection to V16 Simulator Results

V16 established:

- $d_s = 3.008$ (late mean) — 3D attractor confirmed
- Gravity $\mathrm{R}^2 = 0.906$, slope $= 1.24$ using $\Omega_i \cdot \Omega_j / r^2$ probe
- $G_{\text{sim}} / \ln 2 = 0.197$ (scale needs D_0 in SI units)

The CWRC/IDV framework suggests the following extensions for V17:

V17 Extensions from CWRC

Add Φ field (wave layer):

$$\Phi_i \leftarrow \Phi_i + \eta \sum_{j \in \mathcal{N}(i)} D_{\text{KL}}(P_i \| P_j) \cdot \sin(\Phi_j - \Phi_i)$$

Add Ψ_i modes (pattern identity):

$$\Psi_i \leftarrow 0.9\,\Psi_i + 0.1\,\overline{\Psi}_{\mathcal{N}(i)} + \varepsilon_0\,\xi$$

Selective coupling: Replace uniform KL coupling with $\kappa_{ij} = D_{\text{KL}}(P_i \| P_j) \cdot e^{-|\Psi_i - \Psi_j|/\sigma}$

Measure IDV alongside d_s: Track whether $\overline{V^{\text{int}}}$ inflates when global d_s is locked at 3. This would be the first direct test of the marble intuition.

5 What V20.2 Is Showing

V20.2 (The Speciation Epoch) runs all three layers simultaneously (J, Φ, Ψ) but shows $d_s \approx 8$ throughout. This is not a failure. In CRF language:

$d_s \approx 8$ with Ψ speciation occurring means the system is in a high-D_{KL} phase where geometry has not yet crystallised. The Φ wave field is present but not yet coupled to the metric tensor strongly enough to drive $d_s \to 3$.
The fix is: metric coupling $G_{\text{new}} \mathrel{+}= \lambda \cdot \cos(\Phi_j - \Phi_i) \cdot v\,v^\top$ with λ large enough that geometry responds to phase alignment. This is exactly what V20.2 does partially but needs to be strengthened.

6 Status Table

Result	**Status**	**Note**
CWRC = re-expression of δ-chain	Derived	$J = G$, $\Phi = D_{\mathrm{KL}}$, $\Psi = P_M$
Kuramoto from CRF Synchronicity	Derived	Special case of $\Pr(\mathrm{sync}) \propto e^{-\alpha D_{\mathrm{KL}}}$
Speciation = low-D_{KL} clustering	Derived	Sessions 4–5, Collective Dynamics
IDV formula	Hypothesis	Consistent with δ-chain, not formally derived
IDI (internal expansion) observable	Open	Not yet measured in any simulation
V17 with Φ+Ψ+IDV	Open	Builds on V16 canonical base
V20.2 $d_s \to 3$ fix	Open	Needs stronger metric–Φ coupling

The cascade $J \to \Phi \to \Psi_i$ was always there in δ. CWRC is the name we give it when we see it as a wave.